

Automate Checkmk with Ansible



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👋 About me



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- 🧳 Self-employed consultant for:
 - Red Hat Ansible (Automation Platform)
 - Red Hat Enterprise Linux
 - Red Hat Satellite
 - Red Hat Identity Management (IPA)
- 📈 Experienced monitoring user (Nagios, Icinga, Checkmk)



👋 About me



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**Please introduce
yourself**



Timetable



-  09:00 – 10:30: Workshop
-  10:30 – 10:45: Break
-  10:45 – 12:30: Workshop
-  12:30 – 13:45: Lunch break
-  13:45 – 15:30: Workshop
-  15:30 – 15:45: Break
-  15:45 – 17:15: Workshop

-  Prepare the LAB environment
-  Installation of Checkmk server
-  Installation of Checkmk agents
-  Checkmk server configuration
-  Fetch information with lookup plugins
-  Using Checkmk as Ansible inventory
-  Self-healing Checkmk agent with Event-Driven Ansible



Preparing the LAB environment





Preparing the LAB environment



-  Create LAB environment
-  Ansible control node =
Checkmk server =
Checkmk agent
-  Use local connection
-  Configure sudo for privileged access



Preparing the LAB environment



-  Slides and code are available on GitHub:
https://github.com/rk-it-at/checkmk_conference_2026
-  Optional: install and clone the repository:

```
git clone https://github.com/rk-it-at/checkmk_conference_2026.git
```



Preparing the LAB environment



-  Available VMs (if you do not have your own)
-  Passwords: <TODO>



Preparing the LAB environment



Hostname	IP Address	Username	Participant
ansible-ws-1		ansible	
ansible-ws-2		ansible	
ansible-ws-3		ansible	
ansible-ws-4		ansible	
ansible-ws-5		ansible	



Preparing the LAB environment



Hostname	IP Address	Username	Participant
ansible-ws-6		ansible	
ansible-ws-7		ansible	
ansible-ws-8		ansible	
ansible-ws-9		ansible	
ansible-ws-10		ansible	



Preparing the LAB environment



-  Configure sudo for privileged access

```
$ su -  
# <EDITOR> /etc/sudoers.d/ansible
```

```
<USERNAME> ALL=(ALL) NOPASSWD: ALL
```

NOTE

Use the root user to run this command!

Replace <EDITOR> with vi, nano or your favorite editor.

Replace <USERNAME> with your system user.



LAB 1: Configure sudo





LAB 1: Configure sudo



-  Configure sudo for privileged access

```
$ su -  
# <EDITOR> /etc/sudoers.d/ansible
```

```
<USERNAME> ALL=(ALL) NOPASSWD: ALL
```

NOTE

Use the root user to run this command!

Replace <EDITOR> with vi, nano or your favorite editor.

Replace <USERNAME> with your system user.

Install Ansible



-  Most Linux distributions ship two versions:
 - **ansible-core**: Ansible binary + minimal set of collections
 - **ansible**: Ansible and selected collections

- ⚙️ Install Ansible on RHEL 10

```
sudo subscription-manager repos --enable codeready-builder-for-rhel-10-$(arch)-rpms
sudo dnf install https://dl.fedoraproject.org/pub/epel/epel-release-latest-10.noarch.rpm
sudo dnf install ansible-core
```


- ⚙️ Install Ansible on AlmaLinux or Rocky Linux 10

```
sudo dnf config-manager --set-enabled crb
sudo dnf install epel-release
sudo dnf install ansible-core
```

- ⚙️ Install Ansible on openSUSE

```
sudo zypper install ansible
```

- ⚙️ Install Ansible on Ubuntu 24.04+

```
sudo apt update  
sudo apt install ansible
```

- ⚙️ Install Ansible with pip (or consider uv as an alternative)

```
pip install ansible
```

NOTE

Use a virtual environment outside the workshop!



LAB 2: Install Ansible





-  Ensure Ansible is installed

```
ansible --version
```

```
ansible [core 2.21.0]
```

NOTE

The available Ansible version depends on your operating system.
On RHEL 10, use at least version 2.16.18 due to a bug involving PGP keys and the `rpm_key` Ansible module!
On RHEL 8, use at most version 2.16.18!

Prepare Ansible inventory



Prepare Ansible inventory



-  Create a folder for your Ansible code

```
mkdir playbooks
cd playbooks
<EDITOR> hosts
cmk-server ansible_connection=local
cmk-agent ansible_connection=local
```

NOTE

Replace <EDITOR> with vi, nano or your favorite editor.



Test connection to target hosts



-  Make sure your target hosts are reachable

```
ansible all -i hosts -m ping
cmk-server | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.12"
  },
  "changed": false,
  "ping": "pong"
}
cmk-agent | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.12"
  },
  "changed": false,
  "ping": "pong"
}
```

Ansible does not send an ICMP ping to test the connection; it tries to log in to the target machine.



LAB 3: Add target to Ansible inventory





LAB 3: Add target to Ansible inventory



-  Make sure your target hosts are reachable

```
$ ansible all -i hosts -m ping
cmk-server | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
cmk-agent | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
```



Install Checkmk Collection



Install Checkmk Collection



-  Collections provide simplified and consistent content delivery
-  The collection contains:
 - Modules
 - Playbooks
 - Roles
 - Plugins
 - Docs
 - Tests



Install Checkmk Collection



-  Development:
<https://github.com/Checkmk/ansible-collection-checkmk.general>
-  Hosted at Galaxy:
<https://galaxy.ansible.com/ui/repo/published/checkmk/general>
-  Install the collection:

```
ansible-galaxy collection install checkmk.general
```



Installation of Checkmk server



-  Install Checkmk with the provided server role
-  Consider writing your own role to fit your Ansible environment and coding guidelines!
-  Role documentation:
<https://galaxy.ansible.com/ui/repo/published/checkmk/general/content/role/server/>

Installation of Checkmk server



-  Create playbook

```
<EDITOR> cmk_server.yml
```

- name: Install Checkmk server
 - hosts: cmk-server
 - connection: local
 - become: true
- roles:
 - checkmk.general.server

-  Create required variables

```
mkdir host_vars  
<EDITOR> host_vars/cmknserver.yml
```

Installation of Checkmk server



```
checkmk_server_version: "2.5.0p5"
checkmk_server_edition: "community"
checkmk_admin_pw: "AnsibleW0rkshop2026!"
checkmk_server_sites:
  - name: "master"
    version: "{{ checkmk_server_version }}"
    update_conflict_resolution: "abort"
    edition: "{{ checkmk_server_edition }}"
    admin_pw: "{{ checkmk_admin_pw }}"
    state: started
```



LAB 4: Install Checkmk server





LAB 4: Install Checkmk server



-  Ensure Checkmk server is installed

```
ansible-playbook -i hosts cmk_server.yml
```

```
PLAY [Install Checkmk server] *****
```

```
TASK [checkmk.general.server : Validating arguments against arg spec 'main' - Install and manage Checkmk servers] *****
ok: [cmk-server]
```

```
...
```

```
RUNNING HANDLER [checkmk.general.server : Start httpd] *****
ok: [cmk-server]
```

```
PLAY RECAP *****
```

```
cmk-server                : ok=23    changed=11    unreachable=0    failed=0    skipped=25    rescued=0    ignored=0
```

Installation of Checkmk server



-  Ensure HTTPS is configured

```
<EDITOR> cmk_server.yml
```

Installation of Checkmk server



```
...
tasks:
  - name: Install mod_ssl for https access
    ansible.builtin.dnf:
      name: mod_ssl
      state: installed
      notify: Restart httpd

handlers:
  - name: Restart httpd
    ansible.builtin.service:
      name: httpd
      state: restarted
```




LAB 5: Secure Checkmk server





LAB 5: Secure Checkmk server



-  Ensure Checkmk server is listening to HTTPS

```
ansible-playbook -i hosts cmk_server.yml
```

```
...
```

```
PLAY RECAP *****
cmk-server           : ok=20    changed=3    unreachable=0    failed=0    skipped=29    rescued=0    ignored=0
```

-  Access your server: https://<ip_address>/master
-  Username: cmkadmin
-  Password: AnsibleW0rkshop2026!



Installation of Checkmk agent



Installation of Checkmk agent



-  Install the Checkmk agent with the provided agent role
-  Consider writing your own role to fit your Ansible environment and coding guidelines!
-  Role documentation:
<https://galaxy.ansible.com/ui/repo/published/checkmk/general/content/role/agent/>

Installation of Checkmk agent



-  Create playbook

```
<EDITOR> cmk_agent.yml
```

Installation of Checkmk agent



```
---
```

```
- name: Install Checkmk agent
  hosts: cmk-agent
  connection: local
  become: true

  roles:
    - checkmk.general.agent
```

Installation of Checkmk agent



-  Create required variables

```
<EDITOR> host_vars/cmka-agent.yml
```


Installation of Checkmk agent



```
checkmk_agent_version: "2.5.0p5"  
checkmk_agent_edition: "community"  
checkmk_agent_server: localhost  
checkmk_agent_server_validate_certs: "false"  
checkmk_agent_site: "master"  
checkmk_agent_user: "cmkadmin"  
checkmk_agent_pass: "AnsibleW0rkshop2026!"
```



LAB 6: Install Checkmk agent





LAB 6: Install Checkmk agent



-  Ensure Checkmk agent is installed

```
$ ansible-playbook -i hosts cmk_agent.yml
```

```
PLAY [Install Checkmk agent] *****
```

```
...
```

```
PLAY RECAP *****
cmk-agent : ok=19 changed=4 unreachable=0 failed=0 skipped=25 rescued=0 ignored=0
```



Checkmk server configuration





Checkmk server configuration: Folders



-  Create two folders: **/linux** and **/linux/checkmk**
-  Add a **label** to `/linux` and ensure the **Checkmk agent** is used for monitoring

```
<EDITOR> host_vars/cmkn-server.yml
```

```
...
checkmk_folders:
  - path: /linux
    attributes:
      labels:
        os: almalinux
        tag_agent: "cmk-agent"
  - path: /linux/checkmk
```

NOTE

Create the folder in the web UI first and use the API to determine the attribute syntax!

-  Create two folders: **/linux** and **/linux/checkmk**
-  Add a **label** to `/linux` and ensure the **Checkmk agent** is used for monitoring
-  Append a new task to the existing tasks

```
<EDITOR> cmk_server.yml
```




Checkmk configuration: Folders



```
...  
- name: Create folder  
  checkmk.general.folder:  
    name: "{{ item['name'] | default(omit) }}"  
    path: "{{ item['path'] }}"  
    attributes: "{{ item['attributes'] | default(omit) }}"  
    state: present  
    server_url: "https://localhost"  
    site: "master"  
    automation_user: "cmkadmin"  
    automation_secret: "{{ checkmk_admin_pw }}"  
    validate_certs: false  
    notify: "Activate Checkmk changes"  
    loop: "{{ checkmk_folders }}"
```

- + Append a new handler

```
...  
- name: Activate Checkmk changes  
  checkmk.general.activation:  
    server_url: "https://localhost"  
    site: "master"  
    automation_user: "cmkadmin"  
    automation_secret: "{{ checkmk_admin_pw }}"  
    validate_certs: false  
    force_foreign_changes: true  
    run_once: true
```



LAB 7: Checkmk configuration: Folders





LAB 7: Checkmk configuration: Folders



-  Ensure both folders are created

```
ansible-playbook -i hosts cmk_server.yml
```

```
PLAY [Install Checkmk server] *****

TASK [Gathering Facts] *****
ok: [cmk-server]

...

PLAY RECAP *****
cmk-server          : ok=21   changed=3    unreachable=0    failed=0    skipped=29    rescued=0    ignored=0
```



Checkmk server configuration: Hosts



-  Create host **cmk-agent** in folder **/linux/checkmk** with IP address **127.0.0.1**

```
<EDITOR> host_vars/cmk-agent.yml
```

```
...
checkmk_host_settings:
  folder: /linux/checkmk
  attributes:
    ipaddress: "127.0.0.1"
```

NOTE

Create the host in the web UI first and use the API to determine the attribute syntax!

-  Create host **cmk-agent** in folder **/linux/checkmk** with IP address **127.0.0.1**
-  Add a new task

```
$ <EDITOR> cmk_agent.yml
```



```
...
tasks:
  - name: Create host
    checkmk.general.host:
      name: "{{ inventory_hostname }}"
      folder: "{{ checkmk_host_settings['folder'] | default(omit) }}"
      attributes: "{{ checkmk_host_settings['attributes'] | default(omit) }}"
      state: present
      server_url: "https://localhost"
      site: "master"
      automation_user: "cmkadmin"
      automation_secret: "{{ checkmk_agent_pass }}"
      validate_certs: false
      notify: "Activate Checkmk changes"
      loop: "{{ checkmk_host_settings | dict2items }}"
```

- + Append a new handler

```
...
handlers:
- name: Activate Checkmk changes
  checkmk.general.activation:
    server_url: "https://localhost"
    site: "master"
    automation_user: "cmkadmin"
    automation_secret: "{{ checkmk_agent_pass }}"
    validate_certs: false
    force_foreign_changes: true
  run_once: true
```



LAB 8: Checkmk configuration: Hosts





LAB 8: Checkmk configuration: Hosts



-  Ensure host is created

```
$ ansible-playbook -i hosts cmk_agent.yml
```

```
PLAY [Install Checkmk agent] *****

TASK [Gathering Facts] *****
ok: [cmk-agent]

...

PLAY RECAP *****
cmk-agent : ok=21 changed=4 unreachable=0 failed=0 skipped=25 rescued=0 ignored=0
```



Checkmk server configuration: Service discovery



Checkmk configuration: Discovery



-  Run service discovery and add **unmonitored** services
-  Append a task to the existing tasks

```
<EDITOR> cmk_agent.yml
```



Checkmk configuration: Discovery



```
...  
- name: Run service discovery  
  checkmk.general.discovery:  
    host_name: "{{ inventory_hostname }}"  
    state: new  
    server_url: "https://localhost"  
    site: "master"  
    automation_user: "cmkadmin"  
    automation_secret: "{{ checkmk_agent_pass }}"  
    validate_certs: false  
    notify: "Activate Checkmk changes"
```



LAB 9: Checkmk configuration: Service discovery





LAB 9: Checkmk configuration: Discovery



-  Ensure services are monitored

```
ansible-playbook -i hosts cmk_agent.yml
```

```
PLAY [Install Checkmk agent] *****

TASK [Gathering Facts] *****
ok: [cmk-agent]

...

PLAY RECAP *****
cmk-agent : ok=22 changed=4 unreachable=0 failed=0 skipped=25 rescued=0 ignored=0
```





Checkmk server configuration: Users



-  Create user **cmkuser** with email address **cmkuser@localhost**, membership in contact group **Everything** and role **Normal user**

```
<EDITOR> host_vars/cmk-server.yml
```

```
...
checkmk_users:
  - name: cmkuser
    password: "AnsibleW0rkshop2026!"
    email: "cmkuser@localhost"
    fullname: "Checkmk user"
    auth_type: password
    contactgroups:
      - "all"
    roles:
      - "user"
```

-  Create user **cmkuser** with email address **cmkuser@localhost**, membership in contact group **Everything** and role **Normal user**
-  Add a new task after the existing tasks

```
<EDITOR> cmk_server.yml
```



Checkmk configuration: Users



...

- name: Create user

checkmk.general.user:

```
name: "{{ item['name'] }}"
password: "{{ item['password'] | default(omit) }}"
email: "{{ item['email'] | default(omit) }}"
fullname: "{{ item['fullname'] | default(omit) }}"
auth_type: "{{ item['auth_type'] }}"
contactgroups: "{{ item['contactgroups'] | default(omit) }}"
roles: "{{ item['roles'] | default(omit) }}"
state: present
server_url: "https://localhost"
site: "master"
automation_user: "cmkadmin"
automation_secret: "{{ checkmk_admin_pw }}"
validate_certs: false
notify: "Activate Checkmk changes"
loop: "{{ checkmk_users }}"
```



LAB 10: Checkmk configuration: Users





LAB 10: Checkmk configuration: Users



-  Ensure user is created

```
$ ansible-playbook -i hosts cmk_server.yml
```

```
PLAY [Install Checkmk server] *****

TASK [Gathering Facts] *****
ok: [cmk-server]

...

PLAY RECAP *****
cmk-server          : ok=22    changed=3    unreachable=0    failed=0    skipped=29    rescued=0    ignored=0
```



Checkmk server configuration: Downtimes



Checkmk configuration: Downtimes



-  Create a **15-minute** downtime

```
<EDITOR> cmk_downtime.yml
```

Checkmk configuration: Downtimes



```
---

- name: Configure Checkmk downtime
  hosts: cmk-agent
  connection: local
  become: true

  tasks:
    - name: Create downtime
      checkmk.general.downtime:
        host_name: "{{ inventory_hostname }}"
        end_after:
          minutes: 15
        state: present
        server_url: "https://localhost"
        site: "master"
        automation_user: "cmkadmin"
        automation_secret: "{{ checkmk_agent_pass }}"
        validate_certs: false
```



LAB 11: Checkmk configuration: Downtimes



LAB 11: Checkmk configuration: Downtimes



-  Ensure the downtime is created

```
$ ansible-playbook -i hosts cmk_downtime.yml
```

```
PLAY [Configure Checkmk downtime] *****
TASK [Gathering Facts] *****
ok: [cmk-agent]

TASK [Create downtime] *****
changed: [cmk-agent]

PLAY RECAP *****
cmk-agent          : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```



Checkmk server configuration: Rules



-  Create a rule that sets the **Checkmk Agent** service to **warning** when its version differs from the site and to **critical** when **TLS** is disabled

```
<EDITOR> host_vars/cmkn-server.yml
```




Checkmk configuration: Rules



```
...
checkmk_rules:
- ruleset: "checkgroup_parameters:agent_update"
  rule:
    properties:
      description: "Checkmk Agent version"
    location:
      folder: "/linux"
    value_raw: "{ 'agent_version': ('site', {}), 'agent_version_mismatch': 1, 'legacy_pull_mode': 2}"
```

NOTE

Create the rule in the web UI first and use the API to determine the rule value syntax!

-  Create a rule that sets the **Checkmk Agent** service to **warning** when its version differs from the site and to **critical** when **TLS** is disabled
-  Add a task after the existing tasks

```
$ <EDITOR> cmk_server.yml
```



Checkmk configuration: Rules



```
...  
- name: Create rule  
  checkmk.general.rule:  
    rule: "{{ item['rule'] }}"  
    ruleset: "{{ item['ruleset'] }}"  
    state: present  
    server_url: "https://localhost"  
    site: "master"  
    automation_user: "cmkadmin"  
    automation_secret: "{{ checkmk_admin_pw }}"  
    validate_certs: false  
    notify: "Activate Checkmk changes"  
    loop: "{{ checkmk_rules }}"
```



LAB 12: Checkmk configuration: Rules





LAB 12: Checkmk configuration: Rules



-  Ensure the rule for agent monitoring is created

```
ansible-playbook -i hosts cmk_server.yml
```

```
PLAY [Install Checkmk server] *****

TASK [Gathering Facts] *****
ok: [cmk-server]

...

PLAY RECAP *****
cmk-server                : ok=23    changed=3    unreachable=0    failed=0    skipped=29    rescued=0    ignored=0
```



Checkmk server configuration: Passwords



Checkmk configuration: Passwords



-  Create a password for **authenticationtest.com**

```
<EDITOR> host_vars/cmknserver.yml
```

Checkmk configuration: Passwords



```
...
```

```
checkmk_passwords:
```

- name: authenticationtest_com
 title: authenticationtest.com
 password: pass

Checkmk configuration: Passwords



-  Create a password for **authenticationtest.com**
-  Add the task **before the Create rule** task!

```
<EDITOR> cmk_server.yml
```



Checkmk configuration: Passwords



...

```
- name: Create password
  checkmk.general.password:
    name: "{{ item['name'] }}"
    title: "{{ item['title'] | default(omit) }}"
    password: "{{ item['password'] }}"
    state: present
    server_url: "https://localhost"
    site: "master"
    automation_user: "cmkadmin"
    automation_secret: "{{ checkmk_admin_pw }}"
    validate_certs: false
  notify: "Activate Checkmk changes"
  loop: "{{ checkmk_passwords }}"
```

Checkmk configuration: Passwords



-  Create a password for **authenticationtest.com**
-  Add the rule to the existing rules

```
<EDITOR> host_vars/cmk-server.yml
```



```
...
checkmk_rules:
...
- ruleset: "active_checks:httptest"
  rule:
    properties:
      description: "authenticationtest.com"
    location:
      folder: "/"
    value_raw: "{ 'endpoints': [{ 'service_name': { 'prefix': 'auto', 'name': 'HTTPTAuth' }, 'url': 'https://authenticationtest.com/HTTPTAuth/' } ],
      'standard_settings': { 'connection': { 'method': ('get', None), 'auth': ('user_auth', { 'user': 'user', 'password':
        ('cmk_postprocessed', 'stored_password', ('authenticationtest_com', ''))) } } } } }"
    conditions:
      host_name:
        match_on:
          - cmk-agent
        operator: one_of
```

NOTE

Create the rule in the web UI first and use the API to determine the rule value syntax!



LAB 13: Checkmk configuration: Passwords



-  Ensure the rule and password for web monitoring are created

```
$ ansible-playbook -i hosts cmk_server.yml
```

```
PLAY [Install Checkmk server] *****

TASK [Gathering Facts] *****
ok: [cmk-server]

...

PLAY RECAP *****
cmk-server          : ok=24    changed=4    unreachable=0    failed=0    skipped=29    rescued=0    ignored=0
```





**Fetch information
from Checkmk with
lookup plugins**



-  Print the **Checkmk server version**
-  Add a task after the existing tasks

```
<EDITOR> cmk_server.yml
```

...

- name: Print Checkmk server version

ansible.builtin.debug:

```
msg: "Checkmk server version is {{ lookup('checkmk.general.version',  
server_url='https://localhost',  
site='master',  
automation_user='cmkadmin',  
automation_secret=checkmk_admin_pw,  
validate_certs=false)}}
```

```
}}"
```

LAB 14: Checkmk lookups





LAB 14: Checkmk lookups



-  Print Checkmk server version

```
$ ansible-playbook -i hosts cmk_server.yml
```

```
PLAY [Install Checkmk server] *****

TASK [Gathering Facts] *****
ok: [cmk-server]

...

TASK [Print Checkmk server version] *****
ok: [cmk-server] => {
  "msg": "Checkmk server version is 2.5.0p5.community"
}

...

PLAY RECAP *****
cmk-server          : ok=25    changed=4    unreachable=0    failed=0    skipped=29    rescued=0    ignored=0
```



Using Checkmk as Ansible inventory



Checkmk as Ansible inventory



-  Checkmk **should** know about all hosts
-  Checkmk can be a good inventory source for Ansible
-  Grouping hosts usually requires tag configuration

-  Create tags for **testing** and **production** hosts

```
<EDITOR> host_vars/cmik-server.yml
```

...

checkmk_tag_groups:

- name: lifecycle_environment
title: Lifecycle Environment
tags:
 - id: production
title: Production
 - id: testing
title: Testing

-  Create tags for **testing** and **production** hosts
-  Add a task after the existing tasks

```
<EDITOR> cmk_server.yml
```



Checkmk as Ansible inventory



```
...  
- name: Create tag group  
  checkmk.general.tag_group:  
    name: "{{ item['name'] }}"  
    title: "{{ item['title'] }}"  
    tags: "{{ item['tags'] }}"  
    state: present  
    server_url: "https://localhost"  
    site: "master"  
    automation_user: "cmkadmin"  
    automation_secret: "{{ checkmk_admin_pw }}"  
    validate_certs: false  
    notify: "Activate Checkmk changes"  
    loop: "{{ checkmk_tag_groups }}"
```

-  Create an inventory configuration for Checkmk

```
<EDITOR> checkmk.yml
```

NOTE

The inventory plugin filename must end in `checkmk.yml` or `checkmk.yaml` !

```
plugin: checkmk.general.checkmk
server_url: https://localhost
site: master
automation_user: cmkadmin
automation_secret: "AnsibleW0rkshop2026!"
validate_certs: false
groupsources:
  - hosttags
  - sites
```

Preview: Upcoming inventory plugin changes



-  Restrict hosts by folder recursively or non-recursively
-  Exclude hosts based on specific tags
-  Convert hostnames to lowercase
-  Append domains to hostnames
-  In development:

<https://github.com/Checkmk/ansible-collection-checkmk.general/blob/feature/inventory-host-filtering/plugins/inventory/checkmk.py>



LAB 15: Checkmk as Ansible inventory





LAB 15: Checkmk as Ansible inventory



-  Ensure tag groups are created

```
ansible-playbook -i hosts cmk_server.yml
```

```
PLAY [Install Checkmk server] *****

TASK [Gathering Facts] *****
ok: [cmk-server]

...

PLAY RECAP *****
cmk-server                : ok=26   changed=5    unreachable=0    failed=0    skipped=29    rescued=0    ignored=0
```



LAB 15: Checkmk as Ansible inventory



-  Print all hosts and groups

```
ansible-inventory -i checkmk.yml --graph
```

```
@all:
  |--@ungrouped:
  |--@tag_criticality_prod:
  |   |--cmk-agent
  ...
  |--@tag_lifecycle_environment_production:
  |   |--cmk-agent
  |--@tag_lifecycle_environment_testing:
  ...
  |--@tag_lifecycle_environment_production:
  |   |--cmk-agent
  |--@tag_lifecycle_environment_testing:
```




LAB 15: Checkmk as Ansible inventory



-  Print host details

```
ansible-inventory -i checkmk.yml --host cmk-agent
```

```
{
  "checkmk_agent_edition": "cre",
  "checkmk_agent_pass": "AnsibleW0rkshop2026!",
  "checkmk_agent_server": "localhost",
  "checkmk_agent_server_validate_certs": "false",
  "checkmk_agent_site": "master",
  "checkmk_agent_user": "cmkadmin",
  "checkmk_agent_version": "2.5.0p5",
  "checkmk_host_settings": {
    "attributes": {
      "ipaddress": "127.0.0.1"
    },
    "folder": "/linux/checkmk"
  },
  "folder": "/linux/checkmk",
  "ipaddress": "127.0.0.1"
}
```



More checkmk.general modules





More checkmk.general modules



- 🕒 Not covered today due to time constraints
- 👤 `checkmk.general.contact_group`
 - Manage contact groups
- 💻 `checkmk.general.host_group`
 - Manage host groups
- 🧩 `checkmk.general.service_group`
 - Manage service groups
- 📅 `checkmk.general.timeperiod`
 - Manage time periods

More checkmk.general modules

-  `checkmk.general.bakery`
 - Bake and sign Checkmk agents
-  `checkmk.general.site`
 - Create, update and manage Checkmk sites
-  `checkmk.general.dcd`
 - Manage dynamic host management
-  `checkmk.general.ldap`
 - Manage LDAP connections
-  Module documentation:

<https://galaxy.ansible.com/ui/repo/published/checkmk/general/content/>

(Bonus) LAB 16: Self-healing Checkmk agent





LAB 16: Self-healing Checkmk agent



- 🎯 Goal: automatically restart a failed Checkmk agent
- 🔔 Checkmk sends service alerts to an EDA webhook
- 📖 A rulebook matches a critical `Check_MK` service
- ▶️ `run_playbook` starts the remediation playbook
- ♻️ The playbook restarts `check-mk-agent.socket`



LAB 16: Install Event-Driven Ansible



-  Install the rulebook runner

```
pip install ansible-rulebook
```

-  Install OpenJDK

```
sudo dnf install java-latest-openjdk
```

-  Install the EDA collection

```
ansible-galaxy collection install ansible.eda
```



LAB 16: Install Event-Driven Ansible



- 🔍 Verify the installation

```
ansible-rulebook --version
```

```
ansible-rulebook [1.3.0]
Executable location = /home/ansible/.local/bin/ansible-rulebook
Drools_jpy version = 0.4.0
Java home = /usr/lib/jvm/java-latest-openjdk
Java version = 26.0.1
Ansible core version = 2.21.0
Python version = 3.12.13
Python executable = /usr/bin/python3
Platform = Linux-6.12.0-211.20.1.el10_2.x86_64-x86_64-with-glibc2.39
```




LAB 16: Install the notification script



-  Create `notify_eda.sh`

```
<EDITOR> notify_eda.sh
```



LAB 16: Install the notification script



```
#!/usr/bin/env bash
# Event-Driven Ansible

URL="${NOTIFY_PARAMETER_1}"

JSON=$(cat <<EOF
{
  "hostname": "${NOTIFY_HOSTNAME}",
  "hostoutput": "${NOTIFY_HOSTOUTPUT}",
  "hoststate": "${NOTIFY_HOSTSTATE}",
  "servicename": "${NOTIFY_SERVICEDESC}",
  "serviceoutput": "${NOTIFY_SERVICEOUTPUT}",
  "servicestate": "${NOTIFY_SERVICESTATE}",
  "date": "${NOTIFY_SHORTDATETIME}",
  "type": "${NOTIFY_NOTIFICATIONTYPE}",
  "what": "${NOTIFY_WHAT}"
}
EOF
)

curl -X POST -H "Content-Type: application/json" -d "${JSON}" ${URL}
exit $?
```



LAB 16: Install the notification script



-  Deploy notification script to Checkmk server

```
<EDITOR> cmk_server.yml
```



LAB 16: Install the notification script



```
...
- name: Install EDA notification script
  ansible.builtin.copy:
    src: notify_eda.sh
    dest: /omd/sites/master/local/share/check_mk/notifications/notify_eda.sh
    owner: master
    group: master
    mode: "0755"
  notify: Restart Checkmk site

handlers:
- name: Restart Checkmk site
  ansible.builtin.command: omd restart master
...
```



LAB 16: Install the notification script



-  Ensure the notification script is installed

```
ansible-playbook -i hosts cmk_server.yml
```

```
PLAY [Install Checkmk server] *****

TASK [Gathering Facts] *****
ok: [cmk-server]

...

PLAY RECAP *****
cmk-server          : ok=27   changed=7    unreachable=0    failed=0    skipped=29    rescued=0    ignored=0
```



LAB 16: Define the notification rule



-  Create the notification rule **Forward alerts to Event-Driven Ansible** and send notifications to **localhost:5050**

```
<EDITOR> host_vars/cmik-server.yml
```



LAB 16: Define the notification rule



```
checkmk_notifications:
- rule_properties:
  description: "Forward alerts to Event-Driven Ansible"
  notification_method:
    notify_plugin:
      option: "create_notification_with_custom_parameters"
      plugin_params:
        plugin_name: "notify_eda.sh"
        params:
          - "http://127.0.0.1:5050"
      notification_bulking:
        state: "disabled"
    contact_selection:
      members_of_contact_groups:
        value:
          - all
      state: "enabled"
```



LAB 16: Define the notification rule



-  Create the notification rule **Forward alerts to Event-Driven Ansible** and send notifications to **localhost:5050**

```
<EDITOR> cmk_server.yml
```




LAB 16: Create the notification rule



...

- name: Create notification rule
checkmk.general.notification:
 rule_config: "{{ item }}"
 state: present
 server_url: "https://localhost"
 site: "master"
 automation_user: "cmkadmin"
 automation_secret: "{{ checkmk_admin_pw }}"
 validate_certs: false
 notify: "Activate Checkmk changes"
 loop: "{{ checkmk_notifications }}"



LAB 16: Create the notification rule



-  Ensure the notification rule is created

```
ansible-playbook -i hosts cmk_server.yml
```

```
PLAY [Install Checkmk server] *****

TASK [Gathering Facts] *****
ok: [cmk-server]

...

PLAY RECAP *****
cmk-server                : ok=28   changed=6    unreachable=0    failed=0    skipped=29    rescued=0    ignored=0
```



LAB 16: Create the remediation playbook



-  Create a playbook that **restarts the Checkmk agent**

```
<EDITOR> restart_checkmk_agent.yml
```



LAB 16: Create the remediation playbook



- name: Restart Checkmk agent
- hosts: "{{ target_host | default('cmk-agent') }}"
- become: true
- tasks:
 - name: Restart Checkmk agent socket
 - ansible.builtin.systemd_service:
 - name: check-mk-agent.socket
 - state: restarted



LAB 16: Create the rulebook



-  Create a rulebook that **listens on port 5050** and checks for a **failed Checkmk agent**

```
mkdir rulebooks  
<EDITOR> rulebooks/restart_checkmk_agent.yml
```



LAB 16: Create the rulebook



```
---
- name: Restart Checkmk agent
  hosts: all
  gather_facts: false

  sources:
    - name: Listen for Checkmk notifications
      ansible.eda.webhook:
        host: 127.0.0.1
        port: 5050

  rules:
    - name: Restart a failed Checkmk agent
      condition: >-
        event.payload.servicename == "Check_MK" and
        event.payload.servicestate == "CRITICAL"
      action:
        run_playbook:
          name: restart_checkmk_agent.yml
          extra_vars:
            target_host: "{{ event.payload.hostname }}"
```



LAB 16: Run the rulebook



-  Start the rulebook in the first SSH session

```
ansible-rulebook -i hosts -r rulebooks/restart_checkmk_agent.yml --verbose
```

```
2026-06-11 17:35:09,830 - ansible_rulebook.app - INFO - Starting sources
2026-06-11 17:35:09,830 - ansible_rulebook.app - INFO - Starting rules
...
2026-06-11 17:35:10,506 - ansible_rulebook.rule_set_runner - INFO - Waiting for actions on events from Restart Checkmk agent
2026-06-11 17:35:10,506 - ansible_rulebook.rule_set_runner - INFO - Waiting for events, ruleset: Restart Checkmk agent
```

-  Keep the rulebook running while it waits on port **5050** !



LAB 16: Trigger self-healing



-  Stop the Checkmk agent in the second SSH session

```
sudo systemctl stop check-mk-agent.socket
```

-  Wait for the `check_MK` service to become critical
-  Watch the rulebook run `restart_checkmk_agent.yml`
-  Confirm that the socket was restarted



LAB 16: Trigger self-healing



```
...
2026-06-11 17:42:55,255 - ansible_rulebook.rule_set_runner - INFO - Waiting for events, ruleset: Restart Checkmk agent
2026-06-11 17:43:02,511 - aiohttp.access - INFO - 127.0.0.1 [11/Jun/2026:17:43:02 +0200] "POST / HTTP/1.1" 200 152 "-" "curl/8.12.1"

PLAY [Restart Checkmk agent] *****

TASK [Gathering Facts] *****
ok: [cmk-agent]

TASK [Restart Checkmk agent socket] *****
changed: [cmk-agent]

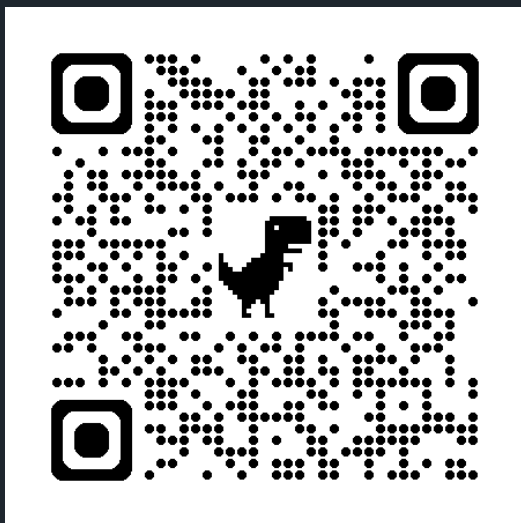
PLAY RECAP *****
cmk-agent          : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```



Feedback



-  <https://forms.gle/ki6KjQF9KS6BavmU8>



🙏 Thank you for participating



- 👤 DI (FH) René Koch
- 📁 Freelancer
- 📅 The Checkmk Conference #12, 18.06.2026